

AI LAB ASSIGNMENT 1

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SEC - CSE 33

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In [1]: import pandas as pd

filename = 'somefile.txt'

with open(filename, 'r', encoding='utf-8') as f:
    lines = [line.strip() for line in f if line.strip()]

data = []
for i in range(0, len(lines), 3):

    if i + 2 < len(lines):
        body = lines[i]
        headline = lines[i+1]
        label = lines[i+2]
        data.append([body, headline, label])

df = pd.DataFrame(data, columns=['News Body Content', 'News Body Headline', 'Label'])

df.to_csv('formatted_data.csv', index=False)

print("Part I Complete: Data saved to 'formatted_data.csv'")
df.head()
```

Part I Complete: Data saved to 'formatted_data.csv'

Out[1]:	News Body Content	News Body Headline	Label
0	A small meteorite crashed into a wooded area i...	Soldier shot, Parliament locked down after gun...	unrelated
1	A small meteorite crashed into a wooded area i...	Tourist dubbed 'Spider Man' after spider burro...	unrelated
2	A small meteorite crashed into a wooded area i...	Luke Somers 'killed in failed rescue attempt i...	unrelated
3	A small meteorite crashed into a wooded area i...	BREAKING: Soldier shot at War Memorial in Ottawa	unrelated
4	A small meteorite crashed into a wooded area i...	Giant 8ft 9in catfish weighing 19 stone caught...	unrelated

```
In [2]: df_words = df.copy()

def count_words(text):
    if pd.isna(text): return 0
    return len(str(text).split())

df_words['Body_Word_Count'] = df_words['News Body Content'].apply(count_words)
df_words['Headline_Word_Count'] = df_words['News Body Headline'].apply(count_words)

output_df_words = df_words[['News Body Content', 'News Body Headline', 'Body_Word_Count', 'Headline_Word_Count']]
output_df_words.to_csv('word_counts.csv', index=False)

print("Part II Complete: Data saved to 'word_counts.csv'")
output_df_words.head()
```

Part II Complete: Data saved to 'word_counts.csv'

```
Out[2]:
```

	News Body Content	News Body Headline	Body_Word_Count	Headline_Word_Count
0	A small meteorite crashed into a wooded area i...	Soldier shot, Parliament locked down after gun...	319	11
1	A small meteorite crashed into a wooded area i...	Tourist dubbed 'Spider Man' after spider burro...	319	11
2	A small meteorite crashed into a wooded area i...	Luke Somers 'killed in failed rescue attempt i...	319	9
3	A small meteorite crashed into a wooded area i...	BREAKING: Soldier shot at War Memorial in Ottawa	319	8
4	A small meteorite crashed into a wooded area i...	Giant 8ft 9in catfish weighing 19 stone caught...	319	24

```
In [3]: import nltk
from nltk.tokenize import word_tokenize

nltk.download('punkt')
nltk.download('punkt_tab')

df_tokens = df.copy()

def count_tokens(text):
    if pd.isna(text): return 0
    return len(word_tokenize(str(text)))

df_tokens['Body-Token-Count'] = df_tokens['News Body Content'].apply(count_tokens)
df_tokens['Headline-Token-Count'] = df_tokens['News Body Headline'].apply(count_tokens)

output_df_tokens = df_tokens[['News Body Content', 'News Body Headline', 'Body-Token-Count', 'Headline-Token-Count']]
output_df_tokens.to_csv('token_counts.csv', index=False)
```

```
print("Part III Complete: Data saved to 'token_counts.csv'")
output_df_tokens.head()
```

```
[nltk_data] Downloading package punkt to
[nltk_data]   C:\Users\KIIT0001\AppData\Roaming\nltk_data...
[nltk_data]   Package punkt is already up-to-date!
[nltk_data] Downloading package punkt_tab to
[nltk_data]   C:\Users\KIIT0001\AppData\Roaming\nltk_data...
[nltk_data]   Package punkt_tab is already up-to-date!
```

Part III Complete: Data saved to 'token_counts.csv'

Out[3]:

	News Body Content	News Body Headline	Body_Token_Count	Headline_Token_Count
0	A small meteorite crashed into a wooded area i...	Soldier shot, Parliament locked down after gun...	364	12
1	A small meteorite crashed into a wooded area i...	Tourist dubbed 'Spider Man' after spider burro...	364	13
2	A small meteorite crashed into a wooded area i...	Luke Somers 'killed in failed rescue attempt i...	364	10
3	A small meteorite crashed into a wooded area i...	BREAKING: Soldier shot at War Memorial in Ottawa	364	9
4	A small meteorite crashed into a wooded area i...	Giant 8ft 9in catfish weighing 19 stone caught...	364	24